

Total No. of Questions : 10]

SEAT No. :

P3231

[5461]-272

[Total No. of Pages : 2

B.E. (Computer Engineering)

SOFTWARE DESIGN METHODOLOGIES AND TESTING

(2012 Pattern) (Semester - II) (End Sem.) (410449)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) What are “include relationship” in use cases? Explain with suitable example and use case diagram. **[5]**

b) Draw a sequence diagram for Online feedback system by a student for a teacher. **[5]**

OR

Q2) a) What is the role designing in any system or project implementation? Explain model based software design and development. **[5]**

b) Explain different degrees of multiplicity of association between classes with suitable example. **[5]**

i) One-to-One Association

ii) Many-to-Many Association.

Q3) a) Explain the real time software architecture with suitable example. **[5]**

b) Explain the intent, motivation, applicability, implementation and consequences of observer pattern? **[5]**

OR

Q4) a) What is singleton pattern? Explain one example scenario where you will singleton pattern to get applied. **[5]**

b) Explain Web Service Protocols in brief. **[5]**

P.T.O.

Q5) a) What is defect management? Discuss the generic steps in defect management process? [8]

b) What is validation? Explain various levels of validation? [8]

OR

Q6) a) Explain V test model with suitable block diagram? State its significance in software development process. [8]

b) Discuss the essential features of modern defect management tool? [8]

Q7) a) Explain top down and bottom integration testing. [6]

b) What is unit testing? List the benefits of unit testing. [6]

c) Briefly explain different types of functional system testing. [6]

OR

Q8) Write short notes on: [18]

a) Performance Testing.

b) Alpha beta Testing.

c) Cyclomatic Complexity.

Q9) a) What are different challenges in mobile testing? Explain MonkeyTalk tool. [6]

b) What is functional testing? Explain its types? [6]

c) What is the purpose of JUnit? [4]

OR

Q10) a) What is automated testing? What are the selection criteria of automated testing tool? [6]

b) What are different components of selenium testing tool? [6]

c) Explain GUI Testing. [4]

